CureGN tranSMART Quick Start Guide
Box Plot with ANOVA

A Box Plot with ANOVA analysis depicts how a categorical variable (e.g. Disease) differs across a continuous variable (e.g. Age).

Step 1: Select samples

Drag and drop terms of interest into the Subset boxes on the **Comparison** tab. Include all terms to be used in the analysis.

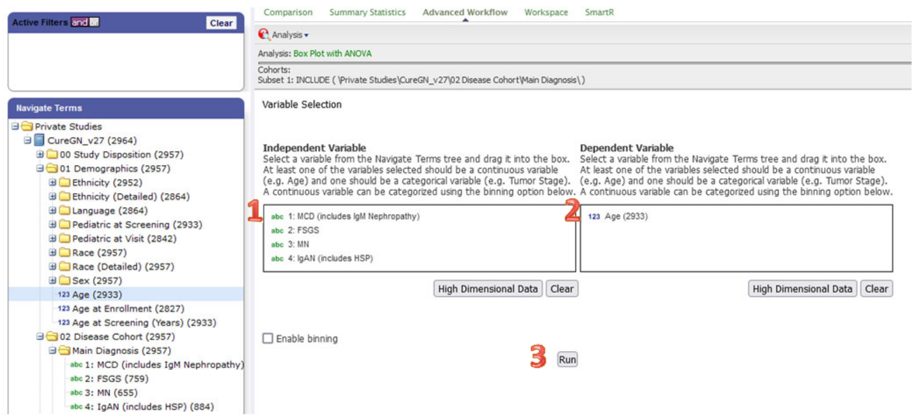
Click on the **Summary Statistics** tab to review the number of samples that will be included in the analysis. Confirm that the set of samples available is sufficient for running a meaningful analysis.

Step 2: Set up analysis

Click on the **Advanced Workflow** tab and select "Box Plot with ANOVA" from the Analysis dropdown list.

Drag terms from the Navigate Terms tree into the Independent Variable (1) and Dependent Variable (2) boxes. You can add multiple terms, or drag over a folder to add all the terms within it. One box should contain a continuous variable (e.g. Age) and the other box should contain a categorical variable (e.g. Disease).

Click the "Run" button (3). A popup box will show a progress bar as the data is compiled and analyzed by tranSMART. Some analyses may take several minutes to run.



Step 3: Review results

Results will appear underneath the variable selection boxes. The Independent variable will appear along the X-axis and the Dependent variable will be the Y-axis (4).

Scroll below the box plot to see ANOVA results (5).

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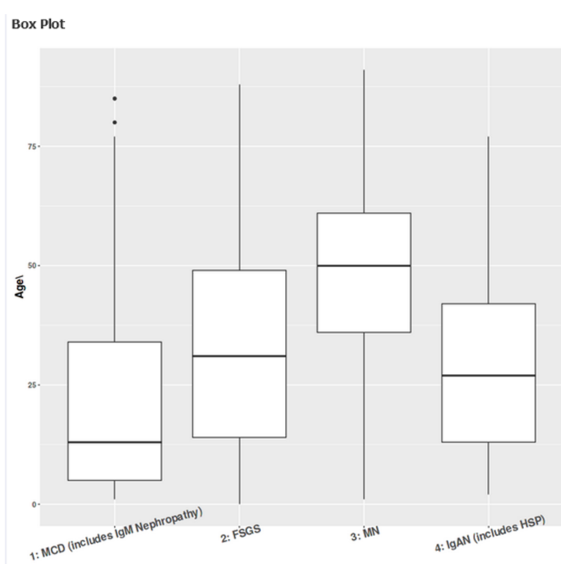

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ANOVA Result

p-value	1.97e-118
F value	201

Group	Mean	n
1: MCD (includes IgM Nephropathy)	21.9	651
2: FSGS	32.6	751
3: MN	47.4	651
4: IgAN (includes HSP)	28.6	880

Pairwise t-Test p-Values

	1: MCD (includes IgM Nephropathy)	2: FSGS	3: MN
2: FSGS	4.10e-24	NA	NA
3: MN	3.84e-112	7.70e-44	NA
4: IgAN (includes HSP)	2.85e-11	4.94e-05	1.39e-72

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